

Contents

Specification Review Report	1
1. Executive Summary	1
2. Overall Maturity	1
3. Findings Summary	2
4. Detailed Findings	2
F-014 — Dedicated Ollama discovery failure test is missing	2
F-015 — Malformed <code>/api/tags</code> item behavior is not evidenced	2
F-016 — Real-model eval failure acceptance test is missing	2
F-017 — Evaluation provenance is thinner than the requested diagnostic goal	3
F-018 — Test identifier ordering is inconsistent	3
5. Requirements Review	3
6. Interface and Data-Contract Review	4
7. State and Failure Review	4
8. Determinism and Algorithm Review	4
9. Edge-Case Review	4
10. Non-Functional Requirement Review	4
11. Security and Trust-Boundary Review	4
12. Observability and Provenance Review	4
13. Testing and Verification Review	4
14. Metrics and Evaluation Review	4
15. Traceability Review	5
16. Internal-Consistency Review	5
17. Architecture Review	5
18. Implementation-Agent Readiness	5
19. Quality Scorecard	5
20. Remediation Plan	5
P0 — Blocking	5
P1 — Important	5
P2 — Improvement	6
21. Final Verdict	6

Specification Review Report

1. Executive Summary

This is a fresh four-pass review of `labs/week4/chapter31/SPEC.md` after adding the PyQt5 UI, Ollama model discovery, levelled diagnostics, robust model-response normalization, natural-language schedule rendering, and the `mortgage eval` harness.

The specification remains coherent and implementation-grade in its deterministic/probabilistic boundary, numerical contracts, CLI surface, GUI model-discovery flow, and evaluation report shape. Five new findings remain: three medium implementation/verification gaps, one medium observability gap, and one low editorial issue. No critical or high-severity blocker was found.

The principal distinction is between **spec completeness** and **implementation conformance**: the spec describes tests T-39, T-43, and T-48 that are not yet present as dedicated automated tests, and the code's Ollama discovery error handling is less defensive than C-08b requires for malformed model entries.

2. Overall Maturity

Level 3 — Implementation-grade, with verification gaps.

A competent engineer can implement the specified system with minimal semantic inference. The remaining findings affect confidence and conformance evidence, not the core architecture.

3. Findings Summary

Severity	Count	IDs
HIGH	0	—
MEDIUM	4	F-014, F-015, F-016, F-017
LOW	1	F-018

4. Detailed Findings

F-014 — Dedicated Ollama discovery failure test is missing

Severity: MEDIUM

Location: §5.2, C-08b, T-43

Observation The specification requires discovery failures to leave a safe dropdown state and display an error, and T-43 names that acceptance test. The current test suite verifies successful `/api/tags` population but does not verify the failure path.

Why it matters A GUI worker failure can regress into a stuck refresh button, an unhandled thread exception, or an empty invalid model selection.

Potential consequence The implementation can claim model discovery support without proving the required degraded behavior.

Recommended resolution Add an offscreen test that makes `OllamaClient.list_models` raise, selects Ollama, waits for the worker to settle, and asserts an error status, a safe dropdown value, and a re-enabled refresh button.

F-015 — Malformed `/api/tags` item behavior is not evidenced

Severity: MEDIUM

Location: C-08b, T-42

Observation C-08b requires malformed JSON or malformed response shape to become `MODEL_ERROR`, but T-42 only tests valid model objects. The implementation also assumes every item supports `.get("name")`.

Why it matters A valid JSON response containing a non-object model item can raise an uncaught `AttributeError` rather than the specified structured failure.

Potential consequence The GUI discovery worker may surface an unclassified exception and fail to restore its controls.

Recommended resolution Specify that `models` MUST be a list of objects and every object MUST contain a string `name`; otherwise return `MODEL_ERROR`. Add malformed-list, missing-name, and non-object-item tests.

F-016 — Real-model eval failure acceptance test is missing

Severity: MEDIUM

Location: C-11, T-48, §5.1 mortgage eval

Observation The spec requires real-model failures to be recorded per case and return exit code 5, but there is no dedicated test for a mock real adapter producing `MODEL_ERROR` during evaluation.

Why it matters The evaluator’s case-failure exit (1) and infrastructure/model-failure exit (5) are materially different automation contracts.

Potential consequence A future change can collapse model outages into ordinary case mismatches or incorrectly pass an interrupted evaluation.

Recommended resolution Add a fake real adapter test that returns `MODEL_ERROR`, asserts the report row outcome/classification, and asserts CLI exit 5.

F-017 — Evaluation provenance is thinner than the requested diagnostic goal

Severity: MEDIUM

Location: C-11, §9.5, §10 observability

Observation The report stores expected/actual values, checks, failure reasons, adapter, and model, but does not store per-case elapsed time, tool-call count, or a bounded raw model-response reference. The CLI’s `DEBUG` stream can show raw payloads, but that data is not retained in `eval_report.json`.

Why it matters A saved report cannot fully explain whether a failure came from slow inference, a missing calculator call, malformed model output, or a wrong extracted field.

Potential consequence Post-run diagnosis requires rerunning the model with `DEBUG` rather than inspecting one durable artifact.

Recommended resolution Add optional per-case `duration_ms`, `tool_calls`, `model_response_excerpt` (bounded and opt-in), and `failure_stage` fields. Keep raw content excluded by default for privacy; `--include-raw` can opt in.

F-018 — Test identifier ordering is inconsistent

Severity: LOW

Location: §9.8

Observation T-44, T-42, and T-43 appear before T-40 and T-41, while the traceability matrix references the identifiers in a different order.

Why it matters This does not change behavior, but it makes review navigation and future additions less predictable.

Potential consequence Reviewers can overlook a test or accidentally reuse an identifier.

Recommended resolution Renumber or reorder the GUI acceptance bullets monotonically in a future documentation pass, preserving existing references through a deliberate ID migration.

5. Requirements Review

Requirements R-01..R-24 are observable and use normative language consistently. R-23 and R-24 correctly capture the newly added model discovery and evaluation surfaces. No requirement contradiction was found.

6. Interface and Data-Contract Review

The canonical request/result/error contracts are strong. C-08b should explicitly constrain every `/api/tags.models[]` item to an object with a string `name`, rather than leaving malformed-item behavior only in prose.

7. State and Failure Review

The GUI worker lifecycle is clear for successful discovery and request execution. The discovery-failure terminal state is specified but lacks automated evidence (F-014). Model-failure versus case-mismatch exit semantics are clear in the CLI table but need T-48.

8. Determinism and Algorithm Review

The four formulas, zero-rate paths, bisection solver, integer tolerance, rounding boundary, and schedule bound remain precise. Evaluation scoring uses explicit tolerances. No algorithmic ambiguity was introduced by the new features.

9. Edge-Case Review

The spec covers unavailable Ollama, malformed JSON, missing fields, aliases, false clarifications, unsupported scope, and model failures. Malformed individual `/api/tags` entries require a concrete contract/test addition (F-015).

10. Non-Functional Requirement Review

Offline mock evaluation, Qt offscreen testing, $O(n)$ amortization, bounded schedules, and worker-thread requirements are measurable. Evaluation timing is logged at runtime but not retained per case, producing the observability gap in F-017.

11. Security and Trust-Boundary Review

The deterministic calculator remains authoritative. Host validation prevents non-HTTP URL schemes, and raw diagnostics are opt-in at DEBUG. If raw model responses are retained in reports, they must be bounded and explicitly opt-in as recommended by F-017.

12. Observability and Provenance Review

CLI diagnostics provide model/phase/payload-size information, and DEBUG provides raw prompts/responses. The durable evaluation report should retain bounded provenance for post-run analysis without making sensitive raw content the default.

13. Testing and Verification Review

The suite covers the deterministic core, GUI success path, model discovery success, response normalization, and mock evaluation. Dedicated tests remain for discovery failure, malformed model-list items, and real-model evaluation failure (F-014..F-016).

14. Metrics and Evaluation Review

The evaluation metrics are defined and reproducible for non-empty case sets. Intent, field, numeric, clarification, and scope accuracy are all computed from per-case checks. The report would be more diagnostically useful with timing/tool-call provenance (F-017).

15. Traceability Review

R-23/R-24 and C-08b/C-11 are present in the traceability matrix. T-43 and T-48 are specified but currently lack dedicated tests, so those rows represent intended verification rather than present evidence.

16. Internal-Consistency Review

The spec is internally consistent. The only editorial issue is non-monotonic GUI test ordering (F-018). The `eval` exit partition correctly distinguishes model failures from ordinary case failures at the specification level.

17. Architecture Review

The architecture remains sound: `cli.py/ui.py` select adapters, `eval.py` compares adapter behavior, `tool.py` delegates to the deterministic core, and Ollama transport remains isolated. No redesign is required.

18. Implementation-Agent Readiness

YES — WITH MINOR CLARIFICATIONS.

A strong coding agent can implement the remaining behavior without material product questions. Before claiming verification-grade conformance, it should add the tests in F-014..F-016 and decide whether to adopt the optional provenance fields in F-017.

19. Quality Scorecard

Dimension	Score
Scope clarity	5
Terminology	4
Requirement precision	4
Interface completeness	4
Data-contract completeness	4
State/lifecycle definition	4
Algorithm precision	5
Failure semantics	4
Edge-case coverage	4
Non-functional requirements	4
Security specification	4
Observability/provenance	3
Testability	4
Evaluation/metrics	4
Traceability	4
Internal consistency	4
Architecture consistency	5
Implementation readiness	4

20. Remediation Plan

P0 — Blocking

None.

P1 — Important

- F-014: Add the GUI discovery-failure test.
- F-015: Pin malformed `/api/tags` item semantics and add tests.

- F-016: Add the real-model evaluation failure test.
- F-017: Decide whether durable per-case provenance is required for operational evaluation.

P2 — Improvement

- F-018: Reorder GUI test identifiers in a future documentation cleanup.

21. Final Verdict

Specification maturity: **Level 3 — Implementation-grade**

Implementation readiness: **READY WITH MINOR FIXES**

Primary blocker: **NONE**

Most important improvement: **Add dedicated failure-path tests for model discovery and real-model evaluation, then strengthen durable evaluation provenance if post-run diagnosis is required.**